Chemico-Biological Interactions 368 (2022) 110222

Contents lists available at ScienceDirect

Chemico-Biological Interactions

journal homepage: www.elsevier.com/locate/chembioint

Research paper

Discovery and characterization of the flavonoids in *Cortex Mori Radicis* as naturally occurring inhibitors against intestinal nitroreductases

Xi Chen [a], Yue Han [a], Lu Chen [a,b], Qian-Lin Tian [a], Yu-Ling Yin [a], Qi Zhou [a], Shi-Zhu Zang [a], Jie Hou [a,c,*]

[a] College of Basic Medical Sciences, Dalian Medical University, Dalian, 116044, China
[b] Institute of Interdisciplinary Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai, 201203, China
[c] State Key Laboratory of Plateau Ecology and Agriculture, Qinghai University, Xining, 810016, China

ARTICLE INFO

Keywords:
Intestinal nitroreductases
Enzyme inhibition
Cortex mori radicis
Flavonoid

ABSTRACT

Gut bacterial nitroreductases are found to be heavily related with the intestinal toxicity of nitroaromatic compounds in food or medicine, which can be converted into mutagenic and enterotoxic nitroso or N-hydroxyl intermediates. Thus, inhibiting the gut microbe-encoded nitroreductases has become an attractive method to reduce the mutagen metabolites in colon and prevent intestinal diseases. In this study, the inhibitory effects of sixteen constituents in *Cortex Mori Radicis* on two kinds of gut bacterial nitroreductases (*Ec*NfsA and *Ec*NfsB) were evaluated with nitrofurazone (NFZ) as substrate and NADPH as electron donor. The results clearly demonstrated that four flavonoids including kuwanon G, kuwanon A, sanggenol A and kuwanon C showed dual inhibition on both *Ec*NfsA and *Ec*NfsB mediated NFZ reduction; morusin, morin, and sanggenone C were strong inhibitors towards *Ec*NfsA; kuwanon H and kuwanon E exhibited effective inhibition on *Ec*NfsB. Further inhibition kinetic analysis and molecular docking simulations displayed that all inhibitors above suppressed both *Ec*NfsA and *Ec*NfsB activities in competitive manners, except non-competitive inhibition of morin on *Ec*NfsA and non-competitive inhibition of kuwanon C on *Ec*NfsB, respectively. Taking together, these findings revealed that most flavonoids in *Cortex Mori Radicis* presented effective inhibition on gut microbial nitroreductases, suggesting that *Cortex Mori Radicis* might be a promising candidate for ameliorating nitroreductases mediated intestinal mutagenicity.

1. Introduction

There are more than 100 trillion microorganisms in the gastrointestinal tract, most of which are found in the colon [1]. For decades, gut microbiome has attracted extensive attention for exploring the host-bacteria interactions in colon diseases [2]. However, most researches always focus on the gut dysbiosis or pathogenic bacteria, gut microbe-encoded enzymes related with the colorectal cancers have been rarely addressed [3,4]. As reported, several enzymes encoded by gut bacteria could produce poisonous metabolites leading to the DNA damage, resulting in the promotion of colon cancer. Among various gut bacterial enzymes, nitroreductases were found to be closely related with carcinogenesis of nitro compounds, which could be reduced to enterotoxic and mutagenic N-nitroso or N-hydroxyl intermediates [5,6]. Moreover, the expression of nitroreductases in feces and intestinal contents of normal samples were much higher than that of colon cancer patients or colon cancer model mice [7,8], leading to the enhancement of the intestinal toxicity. Therefore, inhibition of gut microbial nitroreductases has been considered to be an effective way to protect intestinal epithelium from DNA damage.

Chinese medicinal herbs have been used in a long-term clinical treatment due to its satisfying safety profiles. Considering most herbal medicines are always taken orally, various natural constituents firstly enter into gastrointestine and take effects subsequently. Interestingly, the oral-bioavailability of many natural compounds are generally low, for instance, the absolute oral bioavailability of flavonoid morin was extremely low at 0.45% and plasma concentrations typically <1 μM in humans [9]. However, the low oral bioavailability made flavonoids more active in the intestinal lumen rather than in circulation, providing an explanation for their pharmacological activities [10–12]. Thus, interactions between these active compounds and gut microbial enzymes are increasingly important to unravel the molecular target for herbal

* Corresponding author. College of Basic Medical Sciences, Dalian Medical University, No. 9 West Section Lvshun South Road, 116044 Dalian, China.
E-mail address: houjie@dmu.edu.cn (J. Hou).

https://doi.org/10.1016/j.cbi.2022.110222
Received 3 September 2022; Received in revised form 29 September 2022; Accepted 11 October 2022
Available online 14 October 2022

medicines in recent years. In turn, herbs have also become an available source for screening inhibitors of gut microbial enzymes.

Cortex Mori Radicis, a famous Chinese herbal medicine, has been used for the therapy of many metabolic disorders including hypertension, diabetes, hyperlipidemia, etc [13,14]. In previous studies, we found that active constituents in *Cortex Mori Radicis* could target on different intestinal enzymes to exert pharmacological activities. For instance, sanggenone C, sanggenone D, kuwanon C, kuwanon G could effectively inhibit pancreatic lipase [15], and morin, sanggenon C, kuwanon G, sanggenol A, kuwanon C had effective inhibition on intestinal bacterial β-glucuronidase [9,16]. In this study, the inhibition on gut microbial nitroreductases (*Ec*NfsA & *Ec*NfsB) by 16 main constituents from *Cortex Mori Radicis* were evaluated *via* high-throughput biochemical assays. The results showed that four flavonoids including kuwanon G, kuwanon A, sanggenol A and kuwanon C were dual inhibitors of both *Ec*NfsA and *Ec*NfsB mediated nitrofurazone reduction; morusin, morin, and sanggenone C were effective inhibitors towards *Ec*NfsA; kuwanon H and kuwanon E presented obvious inhibition on *Ec*NfsB. In addition, the ligand binding modes as well as inhibitors-nitroreductases interactions were investigated by molecular docking simulations. The results displayed that all these flavonoid inhibitors suppressed both *Ec*NfsA and *Ec*NfsB activities in competitive manner except morin and kuwanon C, which exhibited non-competitive manner in inhibition of *Ec*NfsA and *Ec*NfsB mediated nitrofurazone reduction, respectively. All these findings revealed that most flavonoids in *Cortex Mori Radicis* presented effective inhibition on gut microbial nitroreductases, which implied that *Cortex Mori Radicis* could be used for relieving intestinal toxicity induced by gut microbial nitroreductases, and also help to deeply understand the

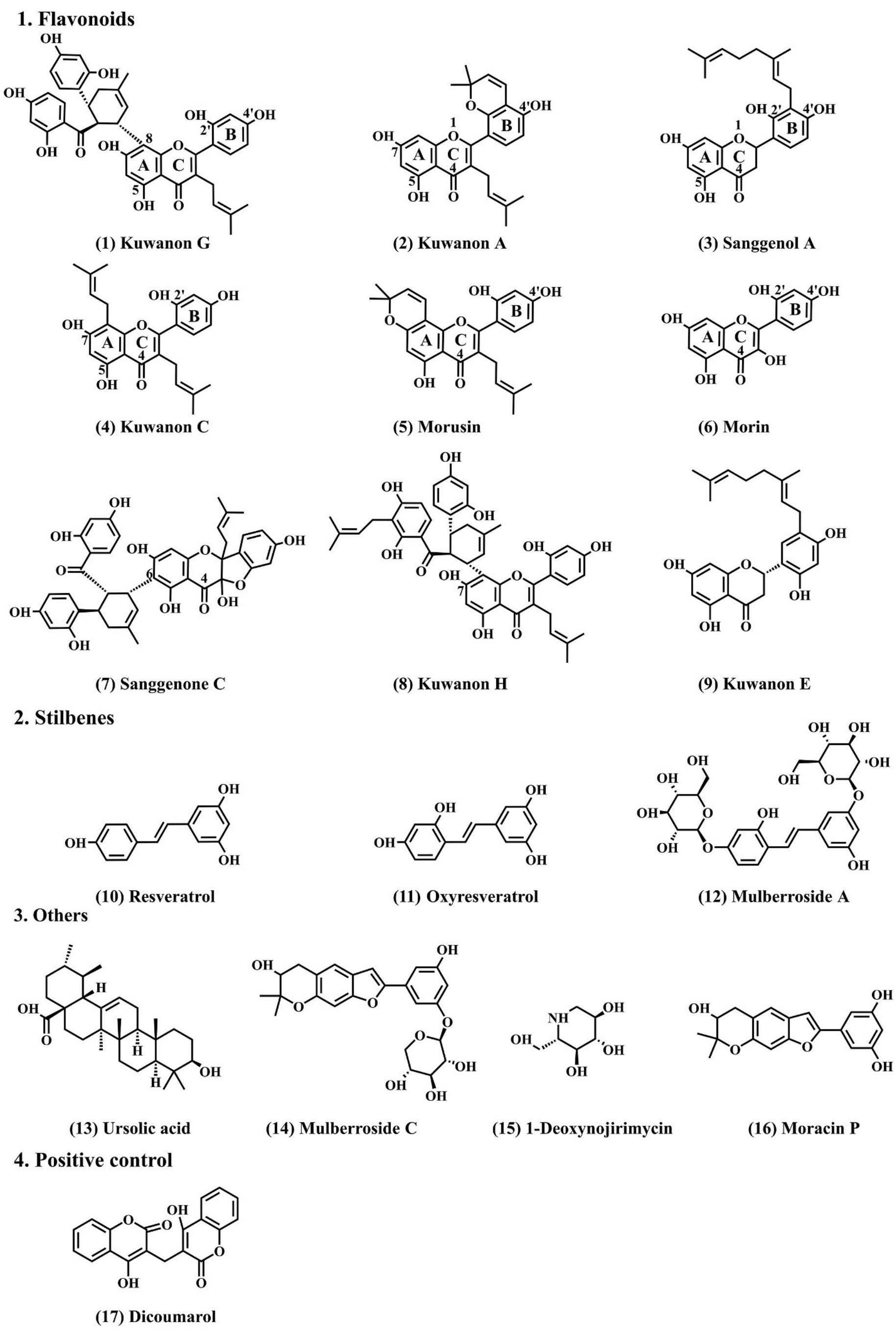

Fig. 1. The major constituents from *Cortex Mori Radicis*.

interactions between natural products in *Cortex Mori Radicis* and gut microbial enzymes.

2. Materials and methods

2.1. Chemicals and reagents

Escherichia coli K12 (MG1655) was purchased from Shanghai Biology Collection Center (Shanghai, China). MiniBEST Bacteria Genomic DNA Extraction Kit Ver.3.0, *Ex Taq*, MiniBEST Agarose Gel DNA Extraction Kit Ver.4.0, DNA Ligation Kit Ver.2.1, *E. coli* competent cells DH5α, NdeI/EcoRI restriction enzymes were purchased from Takara Biomedical Technology Co., Ltd. (Japan). *E. coli* competent cells BL21 (DE3) was ordered from Transgen BioTech Co., Ltd. (Beijing, China). Plasmid pET-28a (+) was purchased from Solarbio Technology Co., Ltd. (Beijing, China). SanPrep Enodotoxin-Free Plasmid Mini Kit was obtained from Sangon Biotech Co., Ltd. (Shanghai, China). Nitrofurazone (NFZ), dicoumarol and NADPH were purchased from MedChemExpress Co., Ltd. (USA). Ni-IDA agarose resin was ordered from Applied Biological Materials Inc. (Canada). Kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)**, kuwanon C **(4)**, kuwanon H **(8)**, kuwanon E **(9)**, mulberroside A **(12)** and mulberroside C **(14)** (purities >98%) were obtained from Baoji Earay Bio-tech Co., Ltd. (Baoji, China). Morusin **(5)**, morin **(6)**, sanggenone C **(7)** and ursolic acid **(13)** (purities >98%) were purchased from Chengdu Pufei De Biotech Co., Ltd. (Chengdu, China). Resveratrol **(10)** and oxyresveratrol **(11)** were supplied by Shanghai Standard Technology Co., Ltd. (Shanghai, China). 1-Deoxynojirimycin **(15)** was gained from Chengdu Biopurify Phytochemicals Ltd. (Chengdu, China). Moracin P **(16)** (purities >98%) was obtained from Baoji Herbest Bio-tech Co., Ltd. (Baoji, China), each test compound was dissolved (100 mM) in DMSO and stored at 4 °C until use (Fig. 1). LC grade DMSO was used in the whole process. Tris-HCl buffer solution (20 mM, pH 7.4) was prepared with millipore water for further use.

2.2. Preparation of Escherichia coli nitroreductases

*Ec*NfsA was prepared and characterized as reported previously by our group [17]. The preparation of *Ec*NfsB was as follows, Bacteria Genomic DNA Extraction Kit was used to extraction and purification of genomic DNA from the culture of *E. coli* K12, according to manufacturer's instructions. *Ex Taq* was used to amplify the *E. coli* nitroreductase NfsB (*Ec*NfsB) target gene from genomic DNA with primers. Forward primer: 5′-GGAATTCCATATGGATATCATTTCTGTCGCCTTAAAG-3' (underline: NdeI); Reverse primer: 5′-GGAATTCTTACACTTCGGTTAAGGTGATGTT-3' (underline: EcoRI). NdeI and EcoRI restriction enzymes were used to digest the target fragment and pET-28a (+) vectors, followed by sticky end connection (Fig. S1). Then, the construct was transformed into *E. coli* DH5α competent cells by heat shock conversion, followed by plating on to LB agar with 50 μg/ml kanamycin. Colonies were picked into LB liquid medium with kanamycin, grown overnight at 37 °C under aerobic conditions, then plasmids were extracted by Enodotoxin-Free Plasmid Mini Kit, according to manufacturer's instructions. After identified by PCR (Fig. S2), the plasmid was transformed into *E. coli* BL21 (DE3) competent cells. The integrity of the insert was finally verified by sequencing. All strains were stored in LB liquid medium with 10% glycerol for long-term storage at −80 °C.

E. coli BL21 (DE3) containing pET-28a (+) with *Ec*NfsB open reading frame was inoculated into LB agar with 50 μg/ml kanamycin and grown overnight with aeration at 37 °C. Colonies were picked into LB liquid medium with kanamycin, shaken at 120 rpm at 37 °C in a temperature-controlled incubator until OD_{600} of 0.4–0.6 was reached. The incubator temperature was reduced to 30 °C then induce with shaking in an Erlenmeyer flask containing 500 μM IPTG for 5 h. Bacterial cultures were pelleted at 8000 rpm, resuspended in phosphate buffered saline (0.1 M, pH 6.8), sonicated in ice bath until suspension viscosity and color were reduced. The suspension was centrifuged at 4 °C and 12,000 rpm for 10 min to obtain a crude enzyme solution. Purification of His-tagged enzyme was performed use an Ni-IDA agarose resins, according to manufacturer's instructions. Fractions contain purified enzyme were verified by SDS-PAGE analysis and stained with Coomassie stain (Fig. S3). The eluent with recombinant enzymes were dialyzed overnight and lyophilized. Finally, the obtained enzyme powder was weighed and storage at −80 °C for further use.

Finally, the properties of both *Ec*NfsA and *Ec*NfsB mediated reduction of NFZ were characterized in Fig. S4, that is, K_m and V_{max} value of *Ec*NfsA were 76.87 ± 5.52 μM and 2.77 ± 0.09 nmol/min/μg protein, as well K_m and V_{max} value of *Ec*NfsB were 78.00 ± 6.94 μM and of 0.56 ± 0.02 nmol/min/μg protein, respectively.

2.3. Enzyme inhibition assays

To explore the inhibitory potential of constituents from *Cortex Mori Radicis* against bacterial nitroreductases, nitrofurazone (NFZ) and dicoumarol were used as the substrate and positive control, respectively. Absorbance of the reaction system by a multi-mode microplate reader (BioTek Synergy H_1, Hybrid Reader, USA) was measured. In detail, the reaction mixture (total volume 200 μl) contained 189 μl Tris-HCl buffer solution (20 mM, pH 7.4), 2 μl enzyme solution (2 μg/ml *Ec*NfsA or 0.6 μg/ml *Ec*NfsB, final concentration), 2 μl NADPH (100 μM, final concentration), 2 μl different concentrations of tested compounds (or DMSO) and 5 μl NFZ (50 μM for *Ec*NfsA or 80 μM for *Ec*NfsB, final concentration). First, the mixture of buffer, enzyme solution, NADPH and inhibitor were pre-incubated under physiological conditions (37 °C). Then, the reaction was started by the addition of the substrate, and absorbance of the reaction system was detected continuously at 400 nm with an interval of 60 s for 30 min. The standard curve of nitrofurazone was shown in Fig. S5. The calculation formula for the residual enzyme activity as follows: the residual enzyme activity (%) = the absorbance in the presence of tested compounds/the absorbance in the negative control (DMSO only) × 100%.

2.4. Inhibition kinetic analysis

In this study, inhibition mechanisms of eight inhibitors, including kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)**, kuwanon C **(4)**, morusin **(5)**, morin **(6)**, sanggenone C **(7)** and kuwanon H **(8)**, against *Ec*NfsA or *Ec*NfsB mediated NFZ reduction were investigated. The reaction rates of NFZ at varying concentrations in the presence of increasing concentrations of each inhibitor were determined. Then, the Lineweaver-Burk plots were drawn to calculate the corresponding inhibition constant (K_i) value. The types of inhibition were determined according to the second plotting of the Lineweaver-Burk plots as previous report [18,19].

2.5. Molecular docking simulations

AutoDockTools 4.2.6 was used to simulate the molecular docking of nitroreductase with inhibitors and substrate. The structure of *Ec*NfsA (PDB ID: 1f5v) and *Ec*NfsB (PDB ID: 1ds7) was downloaded from Protein Data Bank (PDB, http://www.rcsb.org/pdb). The three-dimensional structure of the selected inhibitor was optimized to minimize its energy. After that, automatic tools were used to process the optimized protein structure to remove the water produced by protein crystallization, add polar hydrogens and distribute kollman charges. Considering the different model of inhibition, the grid dimensions containing the active sites of 126 × 70 × 126 Å^3 and 126 × 90 × 126 Å^3 were selected for *Ec*NfsA or *Ec*NfsB docking simulations, respectively. According to Lamarck genetic algorithm, the most favorable prediction confirmation was finally proposed.

2.6. Synergistic or antagonistic effects of inhibitors

According to the previous method [20], four inhibitors including kuwanon G **(1)**, kuwanon C **(4)**, morusin **(5)** and kuwanon H **(8)** were selected to explore the synergistic or antagonistic effects. The dose-dependent response curves IC_{30}, IC_{50} and IC_{70} contour plots were performed, and the combination index (CI) was used to evaluate synergistic and antagonistic effects. CI, the sum of the ratio of inhibitory concentrations, were calculated as follows [17]:

Combination index (CI) $= [D_1 / (D_X)_1] + [D_2 / (D_X)_2]$

$CI < 1$, synergism; $CI = 1$, additive effect; $CI > 1$, antagonistic.

2.7. Statistical analysis

Unless otherwise stated in the full text, all analytical measurements were repeated 3 times (n = 3). GraphPad Prism 8.0 software (California, USA) was used for data processing and analysis. All assays were conducted in triplicate, and the data were shown as mean ± SD.

3. Results

3.1. The inhibitory effects of Cortex Mori Radicis constituents against nitroreductases

Inhibitory effects of sixteen major components including kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)**, kuwanon C **(4)**, morusin **(5)**, morin **(6)**, sanggenone C **(7)**, kuwanon H **(8)**, kuwanon E **(9)**, resveratrol **(10)**, oxyresveratrol **(11)**, mulberroside A **(12)**, ursolic acid **(13)**, mulberroside C **(14)**, 1-deoxynojirimycin **(15)** and morusin P **(16)** (Fig. 1) in *Cortex Mori Radicis* on both *Ec*NfsA and *Ec*NfsB were evaluated with nitrofurazone as the substrate probe and NADPH as the electron donor. The results showed that all tested flavonoids and ursolic acid **(13)** displayed moderate to strong inhibition on *Ec*NfsA or *Ec*NfsB mediated nitrofurazone reduction under concentration of 100 μM, and other kinds of compounds had weak inhibition on both nitroreductases (Fig. 2). Next, the inhibition potentials of nine flavonoids and ursolic acid **(13)** on *Ec*NfsA or *Ec*NfsB were routinely assayed at three different concentrations (1 μM, 10 μM, 100 μM). According to the results, these flavonoids could be divided into three classes: four flavonoids including kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)** and kuwanon C **(4)** as dual inhibitors of both *Ec*NfsA and *Ec*NfsB, morusin **(5)**, morin **(6)** and sanggenone C **(7)** as *Ec*NfsA inhibitors, and kuwanon H **(8)** and kuwanon E **(9)** as *Ec*NfsB inhibitors (Fig. S6). In further, dose-dependence inhibition curves were characterized as shown in Fig. 3 & Fig. 4. Specifically, four flavonoids including kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)** and kuwanon C **(4)** showed effective dual inhibition on NFZ reduction mediated by *Ec*NfsA and *Ec*NfsB, with IC_{50} values of 1.60 ± 0.082 μM and 3.98 ± 0.64 μM, 1.64 ± 0.066 μM and 8.94 ± 1.47 μM, 2.68 ± 0.10 μM and 5.28 ± 0.57 μM, 5.60 ± 0.20 μM and 2.31 ± 0.34 μM, respectively; morusin **(5)**, morin **(6)** and sanggenone C **(7)** could strongly inhibit *Ec*NfsA with IC_{50} values of 1.59 ± 0.065 μM, 3.34 ± 0.16 μM and 3.74 ± 0.38 μM, ursolic acid **(13)** showed moderate inhibition against *Ec*NfsA with IC_{50} values of 15.00 ± 0.45 μM; kuwanon H **(8)** and kuwanon E **(9)** obviously inhibited *Ec*NfsB with IC_{50} values of 1.80 ± 0.24 μM and 22.88 ± 4.68 μM (Table S1). These results indicated that most flavonoids in *Cortex Mori Radicis* had strong inhibitory effects on gut microbial nitroreductases, which prompted us to further study the inhibitory kinetics and molecular mechanisms of these compounds on nitroreductases.

3.2. The kinetics of inhibition of nitroreductase-mediated NFZ reduction

Inhibition mechanisms for these flavonoids were characterized through both time-dependent assays and Lineweaver-Burk plot analysis.

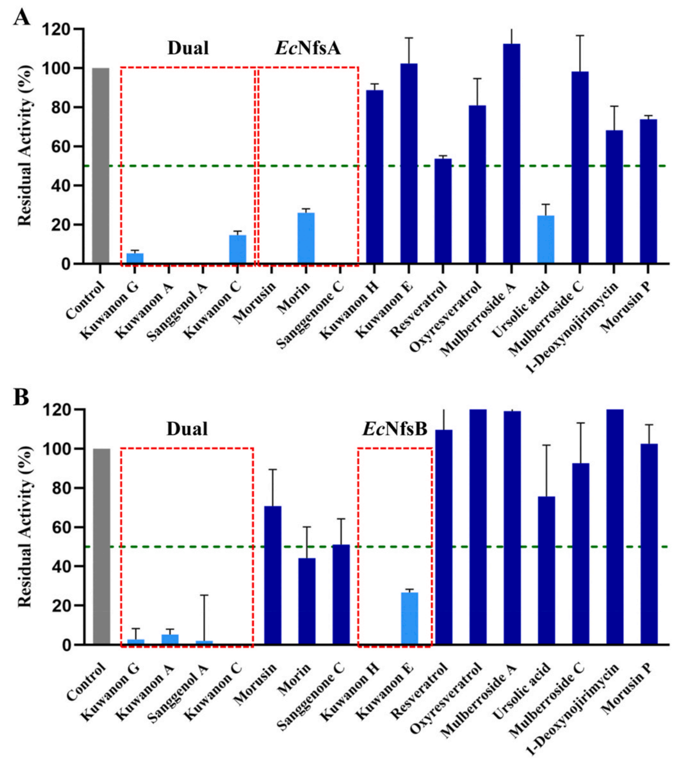

Fig. 2. Inhibitory effect of *Cortex Mori Radicis* components on *Ec*NfsA (A) and *Ec*NfsB (B) at the final concentration of 100 μM. The control group represented DMSO only.

According to Fig. S7 & Fig. S8, all tested flavonoids exhibited time-independent inhibition towards *Ec*NfsA or *Ec*NfsB, indicating that they were reversible inhibitors rather than irreversible inactivators. In further, Lineweaver-Burk plots demonstrated three dual inhibitors consisting of kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)** inhibited both *Ec*NfsA and *Ec*NfsB activities in competitive manners with K_i values of 0.67 ± 0.098 μM and 6.59 ± 1.59 μM, 1.59 ± 0.38 μM and 2.78 ± 1.31 μM, 0.72 ± 0.13 μM and 5.15 ± 0.99 μM, respectively; with one exception, kuwanon C **(4)** suppressed *Ec*NfsA activity in competitive mode with K_i value of 0.53 ± 0.15 μM, but it inhibited *Ec*NfsB activity *via* non-competitive mode with K_i value of 1.39 ± 0.10 μM (Fig. 5 & Fig. 6); *Ec*NfsA inhibitors consisting of morusin **(5)** and sanggenone C **(7)** suppressed the enzyme activity in competitive manners with K_i values of 1.35 ± 0.27 μM and 1.14 ± 0.16 μM, respectively, but morin **(6)** inhibited *Ec*NfsA in a non-competitive manner with K_i value of 2.77 ± 0.25 μM (Fig. S9); as well, kuwanon H **(8)** showed competitive inhibition modes against *Ec*NfsB with K_i values of 1.22 ± 0.21 μM (Fig. S10). Thus, flavonoids in Cortex Mori could precisely targeted on gut bacterial nitroreductases and effectively block activation of nitro-compounds by these enzymes in *vitro* (Table S2 & Table S3).

3.3. Molecular docking simulation

To further investigate the interactions between inhibitors and enzymes, molecular docking simulations were performed with crystal structures of *Ec*NfsA (PDB: 1f5v) and *Ec*NfsB (PDB: 1ds7), respectively. The substrate NFZ made contacts with *Ec*NfsA through a network of hydrogen bonds, involving oxygen atom in nitro group with Arg-225 (2.2 Å and 2.3 Å) and Ser-41 (2.8 Å), as well as the oxygen atom of furan ring with Ser-41 (2.7 Å); NFZ were parallel to FMN with hydrogen bonds (2.1 Å) and Pi-Pi stack (4.2 Å and 4.7 Å) contacts (Fig. S11). Meanwhile, NFZ was docked into *Ec*NfsB with four pairs of hydrogen bonds, that were oxygen atom in carbonyl group with Ser-12 (2.9 Å) and Lys-14 (2.6 Å), nitrogen atom of carbon-nitrogen double bond with Lys-

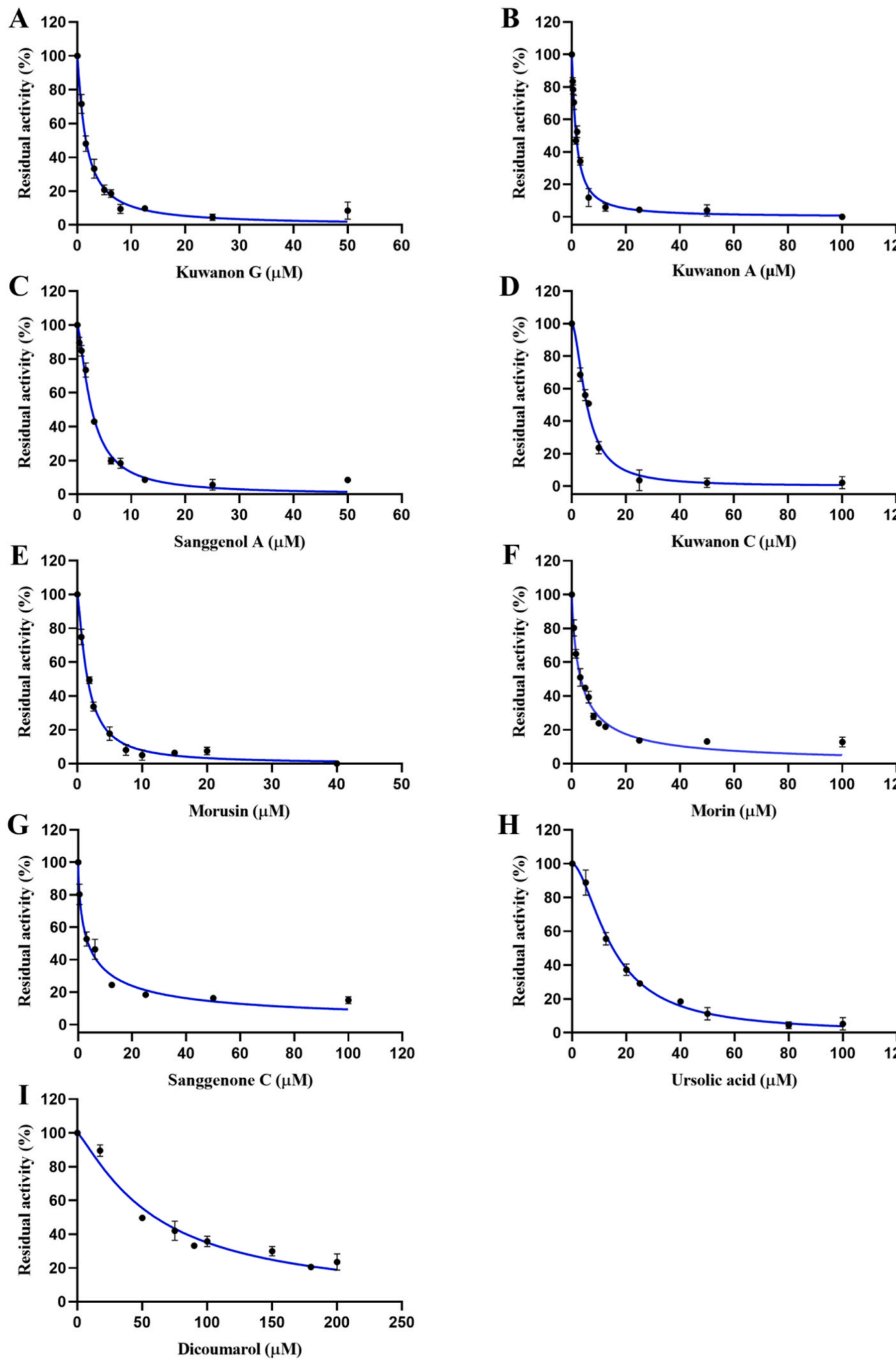

Fig. 3. The dose-response curves of eight compounds in *Cortex Mori Radicis* (A–H) and positive control (I) on *Ec*NfsA. All data were expressed as mean ± SD of triplicate reactions.

14 (2.2 Å) and hydrogen atom in amino group with FMN-218 (2.4 Å and 2.6 Å) (Fig. S12). Thus, the results implied that Arg-225, Ser B-41 and FMN-360 in *Ec*NfsA, as well as Ser-12, Lys-14 and FMN-280 in *Ec*NfsB were key residues in their respective catalytic reactions.

Pairs of hydrogen bonds between inhibitors and *Ec*NfsA played major roles in inhibition reduction of NFZ. The phenolic hydroxyl groups and carbonyl groups of kuwanon G **(1)** can interact with the amino acids Asn-134 (1.9 Å), His-69 (2.4 Å), Gln-67 (2.8 Å) and Gly-65 (2.9 Å), meanwhile, C6–C3 ring can form Pi-Pi stacked interactions with FMN-360 of *Ec*NfsA (Fig. 7A–C). C4′-hydroxyl group and C4-carbonyl group of kuwanon A **(2)** formed hydrogen bonds with Arg-225 (3.0 Å) and Ser-41 (2.8 Å) in the active catalytic site of *Ec*NfsA, and also generated hydrogen bonds with His-215 (3.1 Å), Asn-134 (2.3 Å) and FMN-360 (3.0 Å), hydrophobic interactions such as amide-pi stacked, pi-sigma and pi-alkyl also existed to inhibit NFZ bind to *Ec*NfsA (Fig. 7D–F). Sanggenol A **(3)** created hydrogen bonds between C1–O and C5–OH with the key amino acids Ser-41 (2.6 Å) and Gln-67 (3.0 Å), alkyl interactions among the isopentenyl group and Ile-220, Leu-103 and Leu-43, as well as Pi-sigma bonds between C6–C3 ring and FMN-360 (Fig. 7G–I). The C5–OH and C7–OH of kuwanon C **(4)** interacted with Arg-225 (2.7 Å, 2.8 Å) and Ser-41 (2.5 Å, 2.6 Å) through hydrogen bonds, respectively; the C4-carbonyl group formed a hydrogen bond with Gln-67 (2.9 Å), and C2′-OH formed a hydrogen bond with FMN-360 (2.9 Å), as well as C6–C3 ring also generated Pi-Pi stacked bonds with FMN-360 of *Ec*NfsA (Fig. 7J-L). The C4′-OH and carbonyl group of morusin **(5)** formed hydrogen bonds with Asn-134 (2.0 Å) and Ser-41

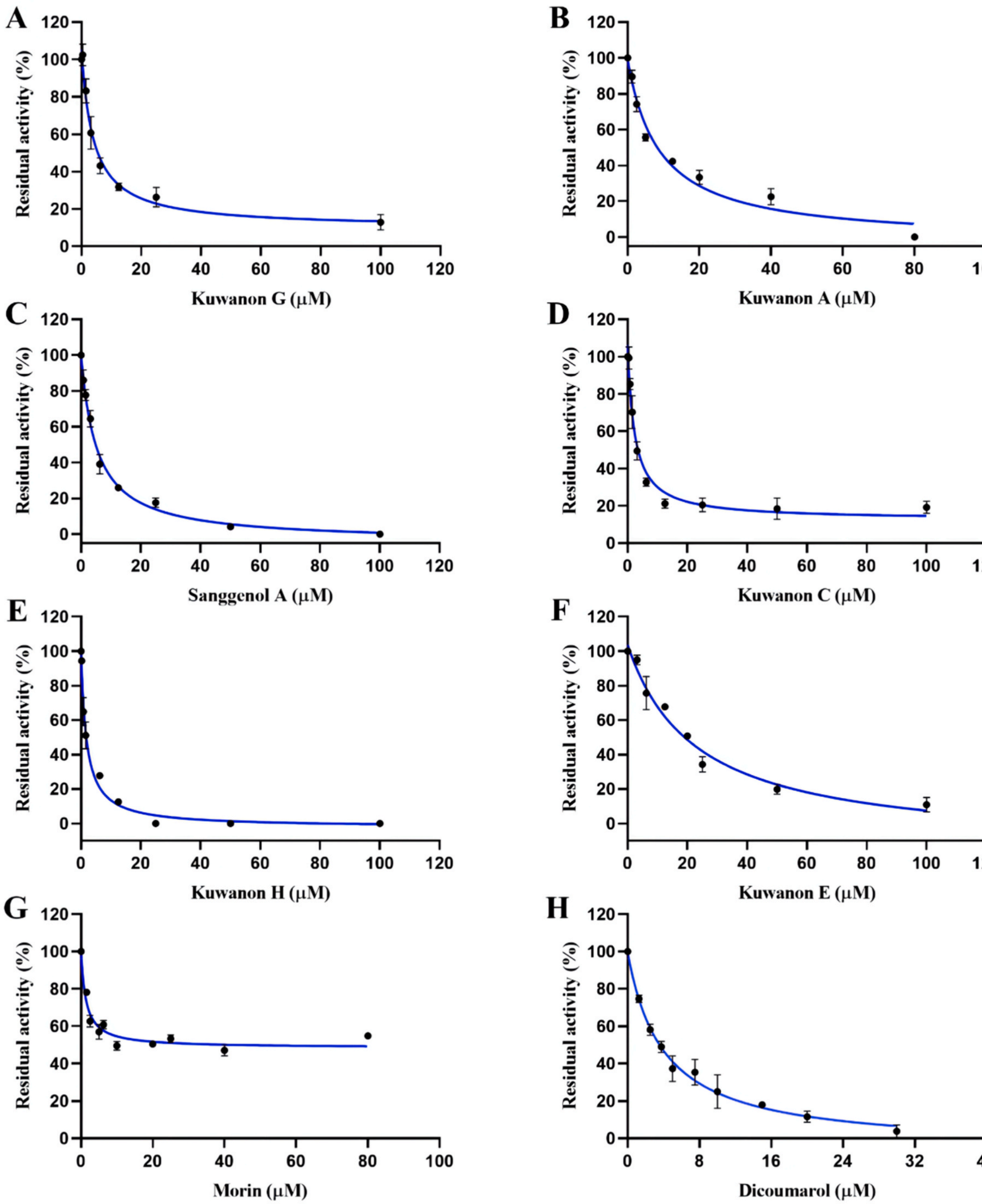

Fig. 4. The dose-response curves of seven compounds in *Cortex Mori Radicis* (A–G) and positive control (H) on *Ec*NfsB. All data were expressed as mean ± SD of triplicate reactions.

(3.1 Å), respectively, the carbonyl group and C5–OH also formed hydrogen bonds with FMN-360 (Figs. S13A–C). In morin **(6)**, carbonyl group, C4′-OH and C6′-OH generated hydrogen bonds with Ser-41 (2.8 Å), Arg-15 (2.6 Å) and Lys-167 (2.3 Å), respectively, and the reason for the non-competitive inhibition with *Ec*NfsA may be due to the relatively small size of this molecule to occupy the active cavity (Figs. S13D–F). As for sanggenone C **(7)**, the phenolic hydroxyl groups formed hydrogen bonds with the Arg-225 (2.7 Å) and Ser-41 (2.7 Å, 3.1 Å) (Figs. S13G–I).

In contrary with *Ec*NfsA, hydrophobic interactions between inhibitors and *Ec*NfsB acted the important role in inhibition NFZ reduction. Kuwanon G **(1)** can well-dock into active cavity of *Ec*NfsB, in which phenolic hydroxyl group and carbonyl group were combined with Arg-121 (2.1 Å), Glu-102 (2.2 Å) and Lys-14 (2.7 Å) by hydrogen bonds, and six hydrophobic interactions including alkyl, Pi-alkyl and Pi-Pi stacked bonds were created among CH3 and Lys-14, C6–C3 ring and Phe-70, as well as benzene ring with Ala-116, respectively (Fig. 8A–C). In kuwanon A **(2)**, O-atom in B-ring, C5–OH, and C7–OH made hydrogen bonds with Lys-14 (2.4 Å), Arg-121 (2.5 Å), Glu-102 (2.5 Å, 2.6 Å) and FMN-218 (2.7 Å), as well as $-CH_3$ in prenyl group at C3 position and C3′ position linked to Phe-70 and Ala-116 through pi-sigma and alkyl interactions, respectively (Fig. 8D–F). Sanggenol A **(3)** formed hydrogen bonds among C2′-OH, C4′-OH, oxygen atom and carboxyl group in C

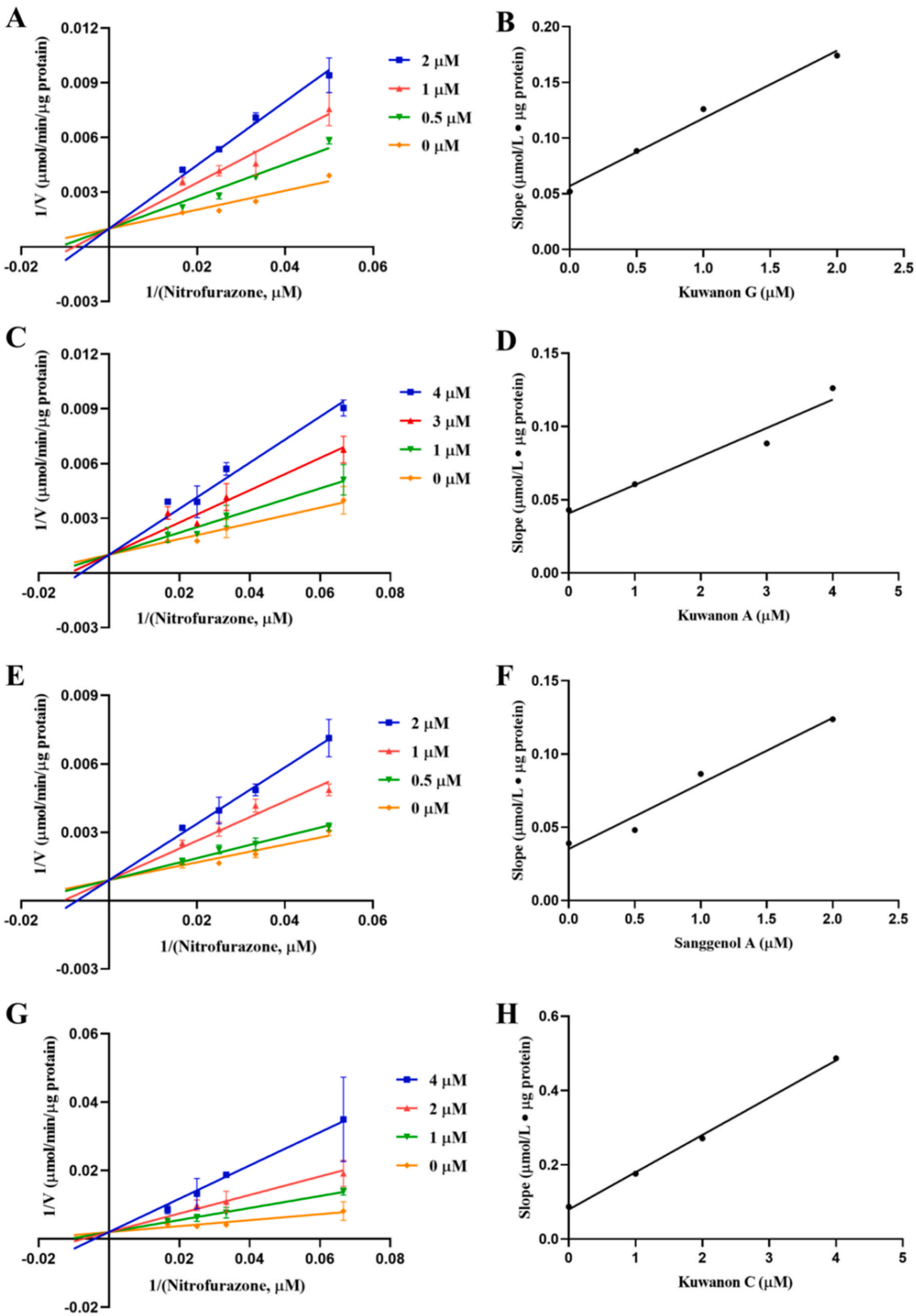

Fig. 5. The inhibitory kinetics of four dual inhibitors in *Cortex Mori Radicis* against *Ec*NfsA-mediated reduction of nitrofurazone. Left: The Lineweaver-Burk plots of kuwanon G (A), kuwanon A (C), sanggenol A (E) and kuwanon C (G). Right: the secondary plots from the Lineweaver-Burk plot for kuwanon G (B), kuwanon A (D), sanggenol A (F) and kuwanon C (H). All inhibition experiments were performed in triplicate and the values were expressed as mean ± SD.

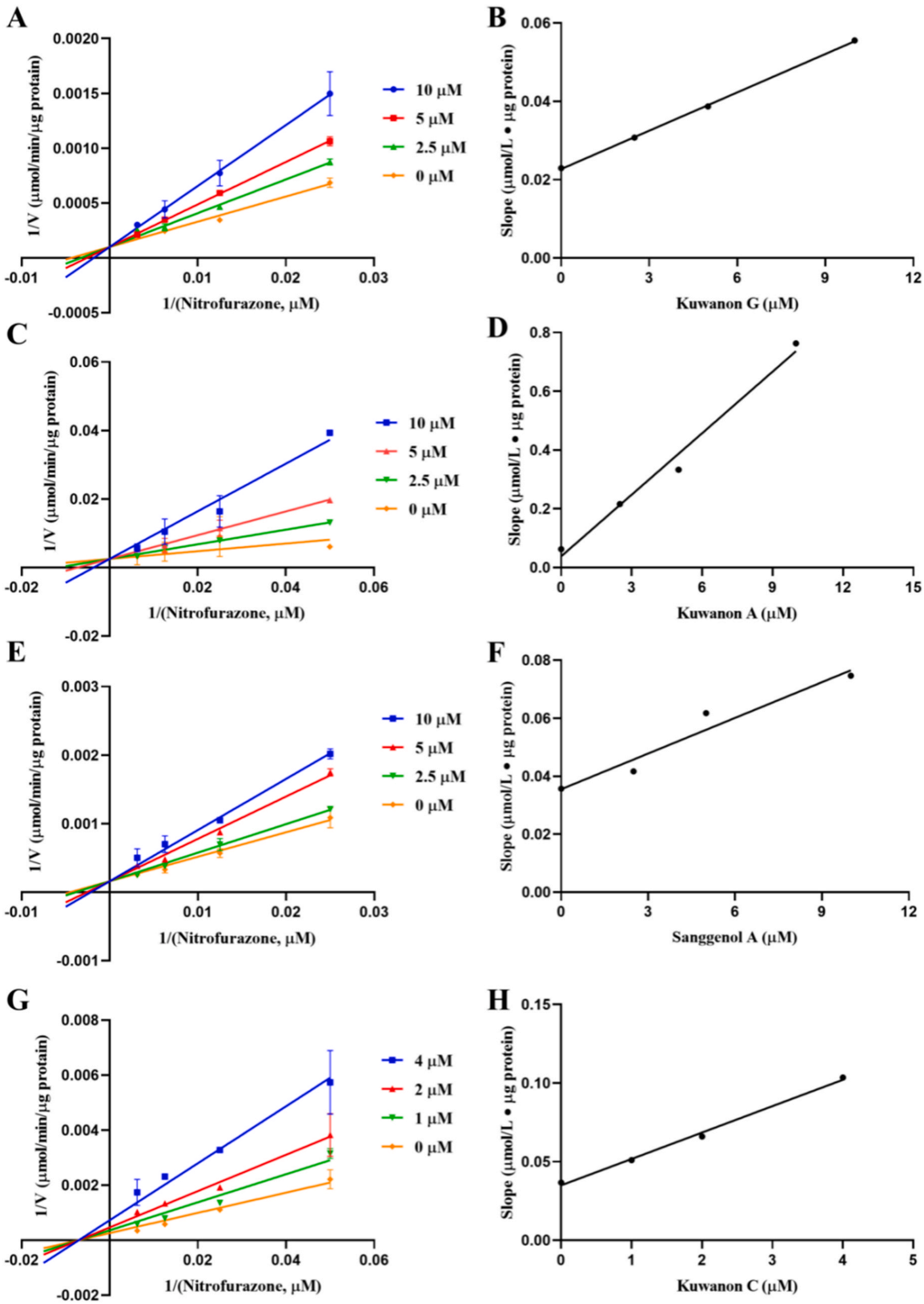

Fig. 6. The inhibitory kinetics of four dual inhibitors in *Cortex Mori Radicis* against *Ec*NfsB-mediated reduction of nitrofurazone. Left: The Lineweaver-Burk plots of kuwanon G (A), kuwanon A (C), sanggenol A (E) and kuwanon C (G). Right: the secondary plots from the Lineweaver-Burk plot for kuwanon G (B), kuwanon A (D), sanggenol A (F) and kuwanon C (H). All inhibition experiments were performed in triplicate and the values were expressed as mean ± SD.

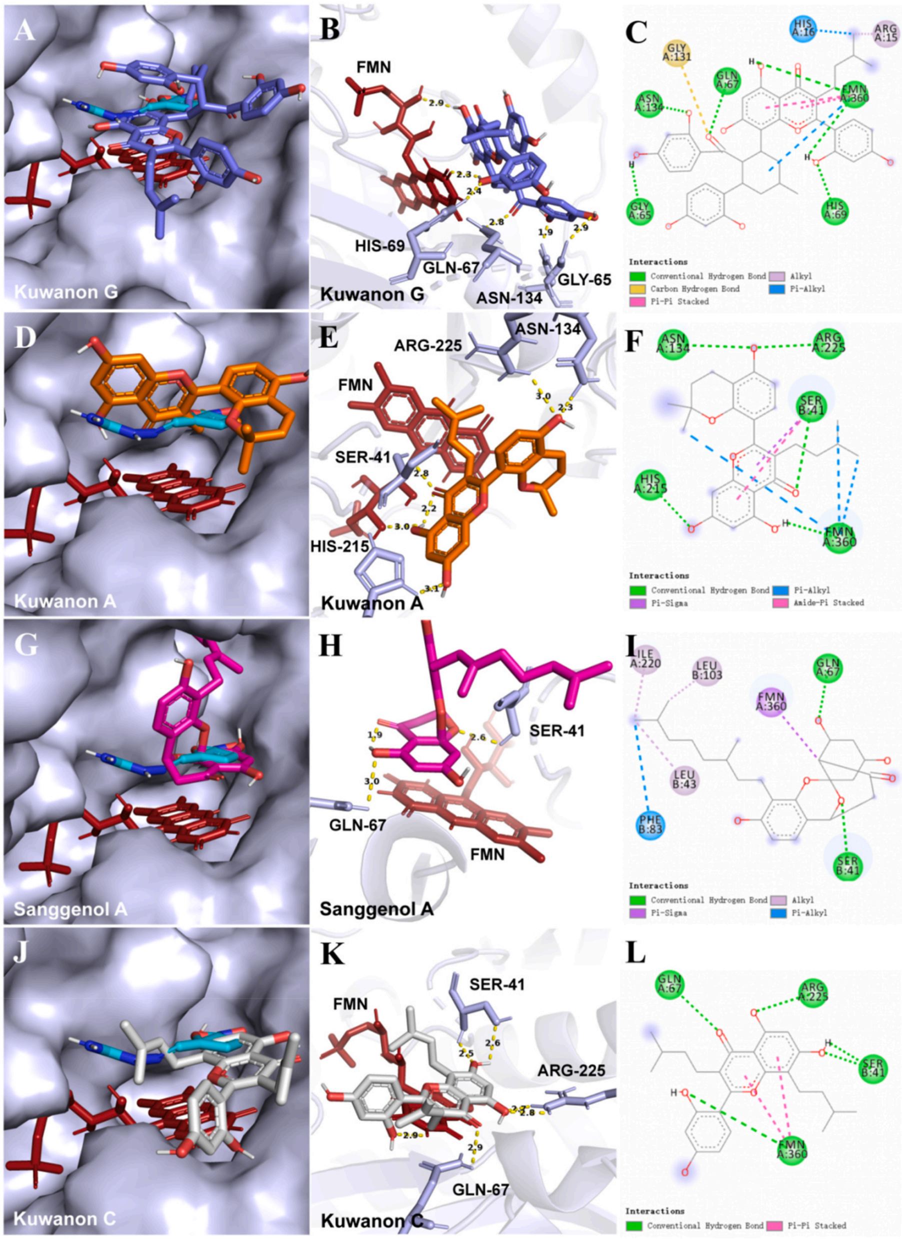

Fig. 7. Stereoscopic views of the structures of *Ec*NfsA (PDB: 1f5v) combined with four dual inhibitors include kuwanon G (A), kuwanon A (D), sanggenol A (G) and kuwanon C (J); Detailed views of the binding region of kuwanon G (B), kuwanon A (E), sanggenol A (H) and kuwanon C (K) to *Ec*NfsA; The 2D interaction of four dual inhibitors with *Ec*NfsA include kuwanon G (C), kuwanon A (F), sanggenol A (I) and kuwanon C (L). (teal: nitrofurazone, purple: kuwanon G, orange: kuwanon A, pink: sanggenol A, grey: kuwanon C, red: FMN, light purple: residues of *Ec*NfsA).

ring with Asn-117 (2.5 Å, 2.8 Å), Arg107 (2.8 Å), and Lys-14 (2.9 Å) of *Ec*NfsB, respectively; while Pi-sigma, Pi-alkyl and alkyl reactions also existed among methyl in prenyl group at C3′ position and Phe-199 or Leu-203 (Fig. 8G–I). In kuwanon C **(4)**, C4 carbonyl group, C5–OH and C2′-OH connected with Asn-117 and FMN218 though hydrogen bonds, and eight pairs of hydrophobic interactions including alkyl, Pi-Pi T-shaped and Pi-alkyl bonds were produced among B-ring or methyl in prenyl groups at C3 or C8 positions with Lys-14, Phe-199, Phe-108, Leu-203, Phe-123, Phe-70 and Phe-124 (Fig. 8J-L). Kuwanon H **(8)** bond to *Ec*NfsB mainly through hydrophobic interactions, such as alkyl, Pi-sigma and Pi-alkyl bonds among methyl groups with Leu 203, Phe-199 and Phe-123, and etc. (Fig. S14). The docking simulations implied that flavonoids in *Cortex Mori Radicis* could inhibit bacterial nitroreductase activities through binding with residues in or near the active cavities of *Ec*NfsA or *Ec*NfsB, and dual inhibitors could produce effective interactions with both enzymes.

3.4. Synergy and antagonism effects of effective inhibitors

In further, the isobolographic plots and interaction index were employed to evaluate synergistic or antagonism effects, for which kuwanon C **(4)** and morusin **(5)** were selected as a pair of *Ec*NfsA inhibitors, and kuwanon G **(1)**, kuwanon C **(4)** and kuwanon H **(8)** were chosen as *Ec*NfsB inhibitors. Dose-response curves of individual inhibitors and dose-pair combined at fixed-ratio concentrations were plotted, respectively. Isobolograms of IC_{30}, IC_{50} and IC_{70} values were depicted, and different dose combinations with specified effect levels were dotted (Fig. S15 & Fig. S16). The combination indexes (CI) were listed in Tables S4–S9. The results showed that kuwanon C **(4)** and morusin **(5)** acted as antagonists of each other in *Ec*NfsA-mediated reduction of NFZ (Fig. 9A); both kuwanon C **(4)** and kuwanon G **(1)** (Fig. 9B), kuwanon C **(4)** and kuwanon H **(8)** (Fig. 9C) acted synergistically on *Ec*NfsB-mediated reduction of NFZ, while kuwanon H **(8)** and kuwanon G **(1)** were antagonists of each other (Fig. 9D). These results were also consistent with the inhibitory modes of these inhibitors.

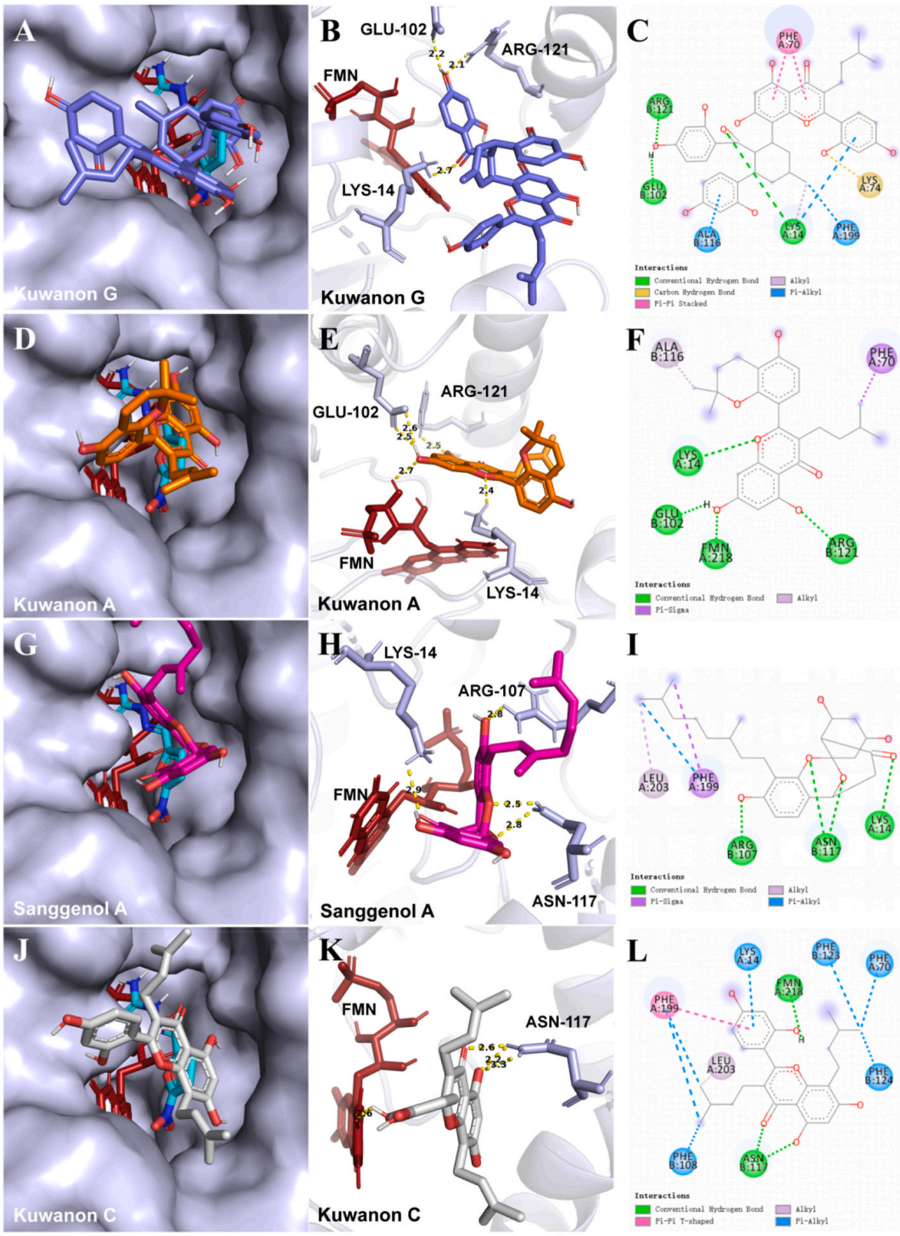

Fig. 8. Stereoscopic views of the structures of *Ec*NfsB (PDB: 1ds7) combined with four dual inhibitors include kuwanon G (A), kuwanon A (D), sanggenol A (G) and kuwanon C (J); Detailed views of the binding region of kuwanon G (B), kuwanon A (E), sanggenol A (H) and kuwanon C (K) to *Ec*NfsB; The 2D interaction of four dual inhibitors with *Ec*NfsB include kuwanon G (C), kuwanon A (F), sanggenol A (I) and kuwanon C (L). (teal: nitrofurazone, purple: kuwanon G, orange: kuwanon A, pink: sanggenol A, grey: kuwanon C, red: FMN, light purple: residues of *Ec*NfsB).

4. Discussion

For decades, gut bacteria imbalance in diseases have attracted much attentions, however, the metabolism enzymes encoded by gut microbiota are still lack of enough understanding. It has been revealed that microbial metabolites are highly related with colon injury [21]. Nitroaromatic compounds in pollutant, food or medicines could be metabolized by microbial nitroreductases in the intestinal tract, in which nitro group is transformed into corresponding hydroxylamine or amino group to generate reactive nitroso or N-hydroxy intermediates. These metabolites can interact with DNA, causing toxic and mutagenic effects [6]. Inhibition of gut bacterial niroreductases is an intriguing approach for preventing colonic DNA damage.

Cortex Mori Radicis is a widely used herb in Asia in the treatment of various acute and chronic diseases. Studies have shown the preventive and therapeutic effects of *Cortex Mori Radicis* on colitis [22,23]. Accordingly, flavonoids with low oral bioactivity accounted for 1.62%–4.55% in *Cortex Mori Radicis*, resulted in more active in the intestinal lumen rather than in circulation. However, the intestinal targets of these active components have not been explored intensively.

Actually, we previously found that main flavonoids including kuwanon G **(1)**, kuwanon C **(4)**, and morusin **(5)**, morin **(6)**, and sanggenon C **(7)** could suppress pancreatic lipase and *Ec*GUS effectively [9,15]. Herein, we assessed the inhibitory effects of 16 major constituents in *Cortex Mori Radicis* on reduction of substrate nitrofurazone by two typical nitroreductases [24–26]. According to the results, it was demonstrated that nine flavonoids in *Cortex Mori Radicis* showed superior inhibition against *E. coli* nitroreductases than the other seven compounds. Interestingly, these nine flavonoids presented different inhibitory effects towards two types of nitroreductases, including four dual inhibitors of *Ec*NfsA and *Ec*NfsB (kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)** and kuwanon C **(4)**), three *Ec*NfsA selective inhibitors (morusin **(5)**, morin **(6)** and sanggenone C **(7)**), and two *Ec*NfsB selective inhibitors (kuwanon H **(8)** and kuwanon E **(9)**). Specifically, the order of inhibition ability (IC_{50} value) of *Ec*NfsA was morusin **(5)** > kuwanon G **(1)** > kuwanon A **(2)** > sanggenol A **(3)** > morin **(6)** > sanggenone C **(7)** > kuwanon C **(4)**; the sequence of inhibition effects on *Ec*NfsB was kuwanon H **(8)** > kuwanon C **(4)** > kuwanon G **(1)** >

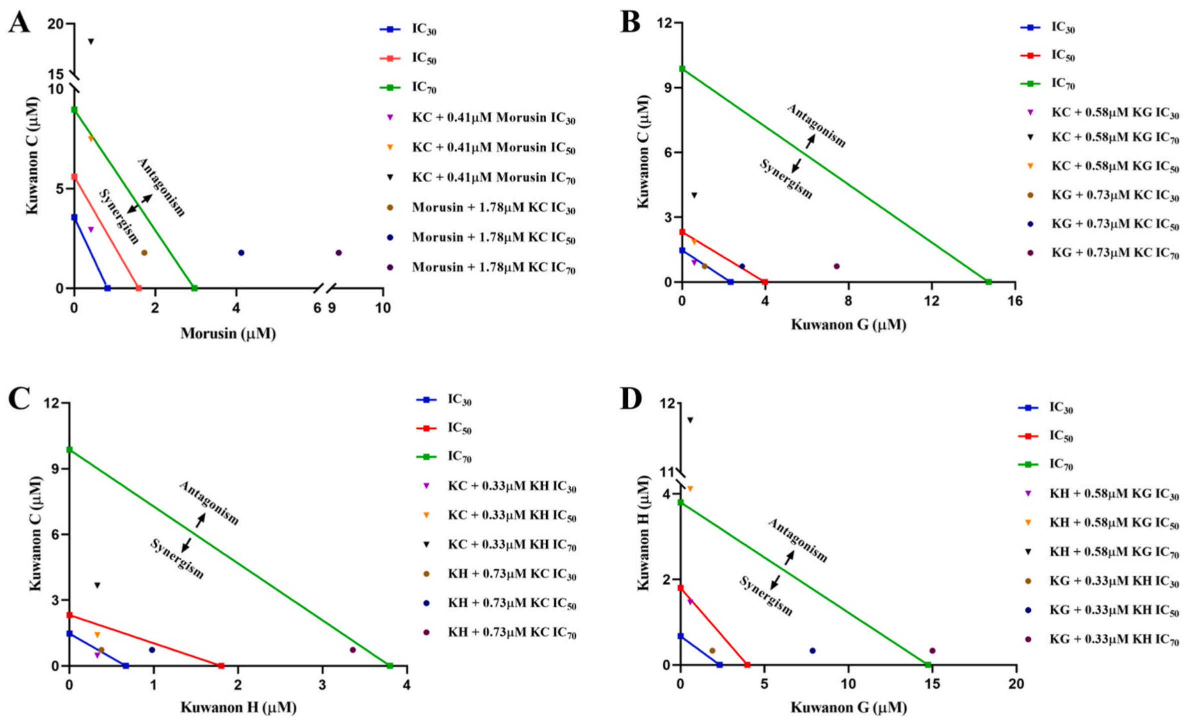

Fig. 9. Isobologram graph of kuwanon C and morusin against *Ec*NfsA-mediated reduction of nitrofurazone (A). Isobologram graph of kuwanon C and kuwanon G (B), kuwanon C and kuwanon H (C), kuwanon H and kuwanon G (D) against *Ec*NfsB-mediated reduction of nitrofurazone.

sanggenol A **(3)** > kuwanon A **(2)** > kuwanon E **(9)**, and most IC_{50} values were lower than 5 μM. Notably, the ethanol extract of *Cortex Mori Radicis* could also effectively inhibit *Ec*NfsA and *Ec*NfsB activities (Fig. S17). Thus, it was strongly implied that flavonoids may play important roles in *Cortex Mori Radicis* inhibition against gut bacterial nitroreductases.

In order to clarify the inhibition mechanisms of these flavonoids towards nitroreductases, lineweaver-Burk plotting was performed. The results revealed that all K_i values were lower than 7 μM, and nine inhibitors played competitive inhibition on *Ec*NfsA or *Ec*NfsB, except kuwanon C **(4)** exerted on *Ec*NfsB and morin **(6)** exerted on *Ec*NfsA *via* non-competitive manners. The results implied that these inhibitors could interact closely with active residues of enzymes.

Despite some biochemical similarities, NfsA and NfsB shared only 10% identity in amino acids sequence with each other, and they belonged to different groups of nitroreductases (Fig. S18) [6]. The substrate NFZ made contacts with *Ec*NfsA through a network of hydrogen bonds, involving oxygen atom in nitro group with Arg-225 (2.2 Å and 2.3 Å) and Ser-41 (2.8 Å), as well as the oxygen atom of furan ring with Ser-41 (2.7 Å); NFZ was parallel to FMN with hydrogen bonds (2.1 Å) and Pi-Pi stack (4.2 Å and 4.7 Å) contacts (Fig. S11). Meanwhile, NFZ was docked into *Ec*NfsB with four pairs of hydrogen bonds, that were oxygen atom in carbonyl group with Ser-12 (2.9 Å) and Lys-14 (2.6 Å), nitrogen atom of carbon-nitrogen double bond with Lys-14 (2.2 Å) and hydrogen atom in amino group with FMN-218 (2.4 Å and 2.6 Å) (Fig. S12). Thus, the results implied that Arg-225, Ser-41 and FMN-360 in *Ec*NfsA, as well as Ser-12, Lys-14 and FMN-280 in *Ec*NfsB were key residues in their respective catalytic reactions. Actually, the docking simulations confirmed that kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)**, kuwanon C **(4)**, morusin **(5)**, and sanggenone C **(7)** bound on *Ec*NfsA at the catalytic sites of nitrofurazone, preventing the interactions between the substrate and the enzyme. It could be clearly seen from Fig. 7 & Fig. S13 that these compounds produced close interactions with FMN, Ser-41, and Arg-225, which were the critical residues for *Ec*NfsA-mediated nitrofurazone reduction (Fig. S11) [27,28]. Hydrogen bonds between hydroxyl groups or carbonyl groups in flavonoids and hydrophilic residues like Ser-41, Gly-65, Asn-134, Arg 225, etc. played a functional role in the competitive inhibition of these flavonoids on *Ec*NfsA. In addition, the positions of substituent hydroxyl groups in flavonoids may be also important for the inhibitory activities, such as morusin **(5)** and morin **(6)** with para-position of OH in C ring were more suitable for blocking *Ec*NfsA activity than quercetin and luteolin with ortho-position OH as previously reported [17].

With regard to *Ec*NfsB, five inhibitors including kuwanon G **(1)**, kuwanon A **(2)**, kuwanon C **(4)**, sanggenol A **(3)**, and kuwanon H **(8)** formed hydrogen bonds with catalytic residues Lys-14 and FMN-218 (Fig. 8 & Fig. S14). But different from *Ec*NfsA, the hydrophobic interactions including Pi-sigma, Pi-alkyl and alkyl bondings were dominant bonds in *Ec*NfsB-inhibitor complexes. Thus, it was implied that four dual inhibitors including kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)** and kuwanon C **(4)** provided both proper hydrophilic interactions for *Ec*NfsA inhibition and hydropholic interactions for *Ec*NfsB inhibition. In further, the kuwanon C **(4)** and morusin **(5)** exhibited strong antagonistic effects towards *Ec*NfsA, also suggesting both two inhibitors had similar binding modes with the enzyme; by contrast, two pair of the *Ec*NfsB inhibitors as kuwanon G **(1)** and kuwanon C **(4)**, kuwanon C **(4)** and kuwanon H **(8)** acted synergistically on *Ec*NfsB-mediated reduction of NFZ, due to the non-competitive inhibition of kuwanon C **(4)** against *Ec*NfsB but competitive inhibition of kuwanon G **(1)** and kuwanon H **(8)** towards *Ec*NfsB. Interestingly, pentenyl structure seemed to be crucial for suppressing activities of both *Ec*NfsA and *Ec*NfsB. It was probably because that isopentenyl flavonoids were more likely to anchor the hydrophobic residues and thus retard the binding of substrate to enzymes. Actually, pentenyl flavonoids had also been found to be effective in inhibition of β-glucuronidases [16], another gut bacteria-encoded enzyme responsible for hydrolysis of drug glucuronides. More and more researches have revealed various bioactivities of these special compounds, which had low oral bioavailability, but great potentials in modulation of gut bacteria-encoded enzymes with broad or specific inhibition abilities.

In previous paper [29], apigenin and tannic acid strongly inhibited the bacterial mutagenesis induced by nitropyrenes, and demonstrated a good correlation between anti-mutagenic effects and their inhibition of nitroreductase activity. In our work, we demonstrated more phenolic compounds—natural flavonoids could effectively inhibit the nitroreductases through dual or selective inhibition. All these results remind us that phenolic compounds with low oral availability may have potential effects on gut bacterial enzymes, which could be the mechanism of various herbal medicines.

Our results showed that the flavonoids in *Cortex Mori Radicis* could effectively inhibit the activity of nitroreductase in different ways. Kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)**, kuwanon C **(4)** exhibited dual inhibition on both *Ec*NfsA and *Ec*NfsB; morusin **(5)**, morin **(6)**, sanggenone C **(7)** were *Ec*NfsA inhibitor; kuwanon H **(8)** and kuwanon E **(9)** only had inhibition on *Ec*NfsB. The inhibition ability of all tested flavonoids exceeded that of the positive inhibitor dicoumarol. It implied that *Cortex Mori Radicis* can reduce the nitro-compounds activation in the intestinal tract by inhibiting the activity of intestinal bacteria nitroreductaes, which also be in agreement that prenylflavonoids could significantly inhibit nitric oxide (NO) production in previous report [30]. In the future, site-directed mutagenesis of amino acid residues should be carried out to further understand its catalytic and inhibitory mechanisms by comparing the changes in catalytic activity. Also, detailed molecular interactions and structure-activity relationships should be explored extensively to explain the inhibition mechanisms of these flavonoids, and to find more effective nitroreductase inhibitors through computer-aided drug design.

5. Conclusions

In summary, the inhibitory effects and mechanisms of the major constituents in *Cortex Mori Radicis* on gut bacterial nitroreductases (*Ec*NfsA & *Ec*NfsB) were carefully investigated for the first time. The results clearly showed that nine major flavonoids isolated from *Cortex Mori Radicis* displayed effective inhibitory effects towards *Ec*NfsA and *Ec*NfsB in different ways. Kuwanon G **(1)**, kuwanon A **(2)**, sanggenol A **(3)** and kuwanon C **(4)** were dual inhibitors against both *Ec*NfsA and *Ec*NfsB, while morusin **(5)**, morin **(6)** and sanggenone C **(7)** effectively suppressed *Ec*NfsA activity but kuwanon H **(8)** and kuwanon E **(9)** showed selective inhibition against *Ec*NfsB. Inhibition kinetic analyses demonstrated that most tested flavonoids blocked nitroreductases-catalyzed nitrofurazone reduction in a competitive manner, while morin **(6)** and kuwanon C **(4)** acted as a non-competitive inhibitor of *Ec*NfsA and *Ec*NfsB, respectively. Molecular docking simulations suggested that these naturally occurring nitroreductases inhibitors could tightly bind on the target enzymes mainly *via* hydrogen bonding and hydrophobic interactions. Collectively, our findings suggested that the flavonoids isolated from *Cortex Mori Radicis* are naturally occurring nitroreductase inhibitors, which offers promising lead compounds for the development of novel agents to reduce the intestinal toxicity of aromatic nitro compounds.

CRediT authorship contribution statement

Xi Chen: Investigation, methodology, validation, data curation, writing - original draft. Yue Han: Methodology, data curation. Lu Chen: Investigation, methodology. Qian-Lin Tian: Investigation. Yu-Ling Yin: Investigation, resources. Qi Zhou: Methodology, resources. Shi-Zhu Zang: Methodology. Jie Hou: Conceptualization, supervision, writing - review & editing, funding acquisition, project administration. All co-authors have seen and agreed on the contents of this manuscript. We warrant that the manuscript has not been submitted to any other journal for simultaneous consideration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was financially supported in part by Natural Science Foundation of China [No. 82160739], State Key Laboratory Project Fund [No. 2022-KF-03], Natural Science Foundation of Liaoning Province [No. 20180530025], Innovative Entrepreneurship Program of High-level Talents in Dalian [2017RQ121] and the Technology Innovation Foundation of Dalian [No. 2020JJ26SN051].

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cbi.2022.110222.

References

[1] C. Jobin, Colorectal cancer: looking for answers in the microbiota, Cancer Discov. 3 (2013) 384–387, https://doi.org/10.1158/2159-8290.CD-13-0042.

[2] I. Chattopadhyay, R. Dhar, K. Pethusamy, A. Seethy, T. Srivastava, R. Sah, J. Sharma, S.A.-O. Karmakar, Exploring the role of gut microbiome in colon cancer, Appl. Biochem. Biotechnol. 193 (2021) 1780–1799, https://doi.org/10.1007/s12010-021-03498-9.

[3] Y. Cheng, Z. Ling, L. Li, The intestinal microbiota and colorectal cancer, Front. Immunol. 11 (2020), 615056, https://doi.org/10.3389/fimmu.2020.615056.

[4] S.L. Gorbach, The intestinal microflora and its colon cancer connection, Infection 10 (1982) 379–384, https://doi.org/10.1007/BF01642307.

[5] D. K R.S. Boddu, O. Perumal, Microbial nitroreductases: a versatile tool for biomedical and environmental applications, Biotechnol. Appl. Biochem. 68 (2021) 1518–1530, https://doi.org/10.1002/bab.2073.

[6] M.D. Roldan, E. Perez-Reinado, F. Castillo, C. Moreno-Vivian, Reduction of polynitroaromatic compounds: the bacterial nitroreductases, FEMS Microbiol. Rev. 32 (2008) 474–500, https://doi.org/10.1111/j.1574-6976.2008.00107.x.

[7] B. Goldin, J. Dwyer, S.L. Gorbach, W. Gordon, L. Swenson, Influence of diet and age on fecal bacterial enzymes, Am. J. Clin. Nutr. 31 (1978) S136–S140, https://doi.org/10.1093/ajcn/31.10.S136.

[8] J. Nakamura, Y. Kubota, M. Miyaoka, T. Saitoh, F. Mizuno, Y. Benno, Comparison of four microbial enzymes in *Clostridia* and *Bacteroides* isolated from human feces, Microbiol. Immunol. 46 (2002) 487–490, https://doi.org/10.1111/j.1348-0421.2002.tb02723.x.

[9] Y. Bai, L. Chen, Y.F. Cao, X.D. Hou, S.N. Jia, Q. Zhou, Y.Q. He, J. Hou, Beta-glucuronidase inhibition by constituents of mulberry bark, Planta Med. 87 (2021) 631–641, https://doi.org/10.1055/a-1402-6431.

[10] G.L. Hostetler, R.A. Ralston, S.J. Schwartz, Flavones: food sources, bioavailability, metabolism, and bioactivity, Adv. Nutr. 8 (2017) 423–435, https://doi.org/10.3945/an.116.012948.

[11] J. Li, Y. Yang, E. Ning, Y. Peng, J. Zhang, Mechanisms of poor oral bioavailability of flavonoid Morin in rats: from physicochemical to biopharmaceutical evaluations, Eur. J. Pharmaceut. Sci. 128 (2019) 290–298, https://doi.org/10.1016/j.ejps.2018.12.011.

[12] S.H. Thilakarathna, H.P. Rupasinghe, Flavonoid bioavailability and attempts for bioavailability enhancement, Nutrients 5 (2013) 3367–3387, https://doi.org/10.3390/nu5093367.

[13] H.J. Eo, G.H. Park, J.B. Jeong, The involvement of cyclin D1 degradation through GSK3 beta-mediated threonine-286 phosphorylation-dependent nuclear export in anti-cancer activity of mulberry root bark extracts, Phytomedicine 23 (2016) 105–113, https://doi.org/10.1016/j.phymed.2015.12.007.

[14] G.Y. Zuo, C.X. Yang, J. Han, Y.Q. Li, G.C. Wang, Synergism of prenylflavonoids from Morus alba root bark against clinical MRSA isolates, Phytomedicine 39 (2018) 93–99, https://doi.org/10.1016/j.phymed.2017.12.023.

[15] X.D. Hou, G.B. Ge, Z.M. Weng, Z.R. Dai, Y.H. Leng, L.L. Ding, L.L. Jin, Y. Yu, Y. F. Cao, J. Hou, Natural constituents from *Cortex Mori Radicis* as new pancreatic lipase inhibitors, Bioorg. Chem. 80 (2018) 577–584, https://doi.org/10.1016/j.bioorg.2018.07.011.

[16] Y. Bai, L. Chen, P.P. Wang, Y.Q. Tang, D.C. Wu, C.L. Zhang, Q. Zhou, R. Yan, J. Hou, Discovery of a naturally occurring broad-spectrum inhibitor against gut bacterial beta-glucuronidases from Ginkgo biloba, Food Funct. 12 (2021) 11190–11201, https://doi.org/10.1039/d1fo01748a.

[17] L. Chen, X. Chen, Y. Bai, Z.-N. Zhao, Y.-F. Cao, L.-K. Liu, T. Jiang, J. Hou, Inhibition of *Escherichia coli* nitroreductase by the constituents in Syzygium aromaticum, Chin. J. Nat. Med. 20 (2022) 506–517, https://doi.org/10.1016/s1875-5364(22)60163-8.

[18] L.H. Wei, T.R. Chen, H.B. Fang, Q. Jin, S.J. Zhang, J. Hou, Y. Yu, T.Y. Dou, Y. F. Cao, W.Z. Guo, G.B. Ge, Natural constituents of St. John's Wort inhibit the

proteolytic activity of human thrombin, Int. J. Biol. Macromol. 134 (2019) 622–630, https://doi.org/10.1016/j.ijbiomac.2019.04.181.
[19] H. Xin, X.Y. Qi, J.J. Wu, X.X. Wang, Y. Li, J.Y. Hong, W. He, W. Xu, G.B. Ge, L. Yang, Assessment of the inhibition potential of Licochalcone A against human UDP-glucuronosyltransferases, Food Chem. Toxicol. 90 (2016) 112–122, https://doi.org/10.1016/j.fct.2016.02.007.
[20] R.J. Tallarida, Revisiting the isobole and related quantitative methods for assessing drug synergism, J. Pharmacol. Exp. Therapeut. 342 (2012) 2–8, https://doi.org/10.1124/jpet.112.193474.
[21] A.E. Lindell, M. Zimmermann-Kogadeeva, K.R. Patil, Multimodal interactions of drugs, natural compounds and pollutants with the gut microbiota, Nat. Rev. Microbiol. 20 (2022) 431–443, https://doi.org/10.1038/s41579-022-00681-5.
[22] M.K. Sriwastva, Z.B. Deng, B. Wang, Y. Teng, A. Kumar, K. Sundaram, J. Mu, C. Lei, G.W. Dryden, F. Xu, L. Zhang, J. Yan, X. Zhang, J.W. Park, M.L. Merchant, N. K. Egilmez, H.G. Zhang, Exosome-like nanoparticles from Mulberry bark prevent DSS-induced colitis via the AhR/COPS8 pathway, EMBO Rep. 23 (2022), e53365, https://doi.org/10.15252/embr.202153365.
[23] Z. Vochyanova, M. Pokorna, D. Rotrekl, V. Smekal, P. Fictum, P. Suchy, J. Gajdziok, K. Smejkal, J. Hosek, Prenylated flavonoid morusin protects against TNBS-induced colitis in rats, PLoS One 12 (2017), e0182464, https://doi.org/10.1371/journal.pone.0182464.
[24] E. Akiva, J.N. Copp, N. Tokuriki, P.C. Babbitt, Evolutionary and molecular foundations of multiple contemporary functions of the nitroreductase superfamily, Proc. Natl. Acad. Sci. U. S. A. 114 (2017) E9549–E9558, https://doi.org/10.1073/pnas.1706849114.
[25] W.A. Denny, Nitroreductase-based GDEPT, Curr. Pharmaceut. Des. 8 (2002) 1349–1361, https://doi.org/10.2174/1381612023394584.
[26] E.I. Hyde, A.K. Chau, L.J. Smith, Backbone assignment of *E. coli* NfsB and the effects of addition of the cofactor analogue nicotinic acid, Biomol NMR Assign. 15 (2021) 143–151, https://doi.org/10.1007/s12104-020-09997-w.
[27] J.N. Copp, A.M. Mowday, E.M. Williams, C.P. Guise, A. Ashoorzadeh, A. V. Sharrock, J.U. Flanagan, J.B. Smaill, A.V. Patterson, D.F. Ackerley, Engineering a multifunctional nitroreductase for improved activation of prodrugs and PET probes for cancer gene therapy, Cell. Chem. Biol. 24 (2017) 391–403, https://doi.org/10.1016/j.chembiol.2017.02.005.
[28] P.R. Race, A.L. Lovering, R.M. Green, A. Ossor, S.A. White, P.F. Searle, C. J. Wrighton, E.I. Hyde, Structural and mechanistic studies of *Escherichia coli* nitroreductase with the antibiotic nitrofurazone. Reversed binding orientations in different redox states of the enzyme, J. Biol. Chem. 280 (2005) 13256–13264, https://doi.org/10.1074/jbc.M409652200.
[29] M.L. Kuo, K.C. Lee, J.K. Lin, Genotoxicities of nitropyrenes and their modulation by apigenin, tannic acid, ellagic acid and indole-3-carbinol in the *Salmonella* and CHO systems, Mutat. Res. 270 (1992) 87–95, https://doi.org/10.1016/0027-5107(92)90119-m.
[30] Z.G. Yang, K. Matsuzaki, S. Takamatsu, S. Kitanaka, Inhibitory effects of constituents from Morus alba var. multicaulis on differentiation of 3T3-L1 cells and nitric oxide production in RAW264.7 cells, Molecules 16 (2011) 6010–6022, https://doi.org/10.3390/molecules16076010.

Update

Chemico-Biological Interactions
Volume 373, Issue , 1 March 2023, Page

DOI: https://doi.org/10.1016/j.cbi.2023.110403

Chemico-Biological Interactions 373 (2023) 110403

Contents lists available at ScienceDirect

Chemico-Biological Interactions

journal homepage: www.elsevier.com/locate/chembioint

Corrigendum to “Discovery and characterization of the flavonoids in Cortex Mori Radicis as naturally occurring inhibitors against intestinal nitroreductases” [Chem. Biol. Interact. 368 (2022) 110222]

Xi Chen [a], Yue Han [a], Lu Chen [a,b], Qian-Lin Tian [a], Yu-Ling Yin [a], Qi Zhou [a], Shi-Zhu Zang [a,**], Jie Hou [a,c,*]

[a] College of Basic Medical Sciences, Dalian Medical University, Dalian, 116044, China
[b] Institute of Interdisciplinary Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai, 201203, China
[c] State Key Laboratory of Plateau Ecology and Agriculture, Qinghai University, Xining, 810016, China

The authors regret < to amend the author Shi-Zhu Zang to the co-corresponding author as following: Xi Chen [a], Yue Han [a], Lu Chen [a,b], Qian-Lin Tian [a], Yu-Ling Yin [a], Qi Zhou [a], Shi-Zhu Zang [a**], Jie Hou [a,c,*] >.

The authors would like to apologise for any inconvenience caused.

DOI of original article: https://doi.org/10.1016/j.cbi.2022.110222.

* Corresponding author. College of Basic Medical Sciences, Dalian Medical University, No. 9 West Section Lvshun South Road, 116044, Dalian, People’s Republic of China.
** Corresponding author. College of Basic Medical Sciences, Dalian Medical University, No. 9 West Section Lvshun South Road, 116044, Dalian, People’s Republic of China.
E-mail addresses: zangsz@dmu.edu.cn (S.-Z. Zang), houjie@dmu.edu.cn (J. Hou).

https://doi.org/10.1016/j.cbi.2023.110403

Available online 14 February 2023